

Evolution Outperforms Random Search but Not Cost-Complexity Pruning:

A Pre-Registered Evaluation of Multi-Objective Evolutionary Decision-Tree Induction

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Abstract

Evolutionary algorithms are a long-standing alternative to greedy induction of decision trees, and multi-objective variants promise a whole accuracy–size trade-off from one run. The system studied here had been reported to find smaller trees than CART at “equivalent” accuracy. We tested that claim under a pre-registered protocol with binding kill criteria: 20 OpenML-CC18 datasets, nested cross-validation, hypervolume over (test accuracy, node count), a random-search baseline matched to the genetic algorithm’s evaluation count exactly, and CART’s full cost-complexity pruning path as the comparator. Two of three hypotheses were rejected. NSGA-II recovers a better frontier than budget-matched random sampling of the same tree space (mean $\Delta\text{HV} +0.66$, Holm-adjusted $p = 0.032$, better on 15 of 20 datasets), so the evolutionary machinery contributes. But it has the larger hypervolume than CART’s pruning path on only 45% of datasets, below the pre-registered 60% threshold, and its tuned single tree loses more than two accuracy points to tuned CART on 8 of 20 datasets, so no equivalence claim can be made. The losses concentrate on datasets where accuracy keeps rising with tree size: the evolved frontiers stop at a median of 5.4 nodes against CART’s 32.0, and an ablation shows that neither the initial bias toward small trees nor the budget alone explains the truncation. The same datasets account for every loss in both runs. Robustness checks—constraint repair, a depth-capped CART path and an exact sparse-tree solver—leave the verdicts unchanged. We also report what the audit behind the study found: published figures that no run had produced, and two evaluation-harness errors of opposite sign, each of which reversed a kill-criterion verdict. We argue that budget-matched random search and full pruning paths should be standard comparators for evolutionary tree induction.

Keywords: decision trees; evolutionary multi-objective optimisation; NSGA-II; benchmarking; pre-registration; negative results; reproducibility.

1 Introduction

Decision trees are among the few model classes whose predictions a person can follow step by step, which keeps them in use wherever a model has to be inspected, explained or signed off (Rudin, 2019). Almost all trees in practice are grown greedily: CART (Breiman et al., 1984) chooses, at each node, the split that most reduces an impurity measure, and controls size afterwards by cost-complexity pruning. Greedy induction optimises a criterion that decomposes over nodes and yields one tree per setting of its pruning parameter.

Evolutionary tree induction instead scores whole trees (Barros et al., 2012; Kretowski, 2019; Rivera-Lopez et al., 2022). With a multi-objective algorithm such as NSGA-II (Deb et al., 2002), it

returns a set of trees spread along the accuracy–size trade-off from a single run (Zhao, 2007), and it accepts objectives that do not decompose over nodes. The usual argument for it is that a global search over whole trees should find models greedy induction misses—as small or smaller at the same accuracy. The comparisons surveyed by Barros et al. (2012) use greedy inducers such as C4.5 and CART as baselines, mostly on UCI datasets, and many report similar predictive performance with smaller trees.

Testing that argument properly needs two further baselines. The first is *random search over the same tree space with the same number of fitness evaluations* (Bergstra and Bengio, 2012). Without it, a favourable result cannot be attributed to evolution rather than to the representation, the fitness function or the sampling distribution of the initial trees. The second is CART’s *entire* cost-complexity pruning path rather than a single tuned or depth-limited tree. A single CART tree is one point on the trade-off; comparing a small evolved tree with it on accuracy compares two points at different sizes, not two methods.

This paper tests one such system—ours—against both baselines. It is, in part, a negative-results paper, and its origin bears on how it should be read. The project’s public materials claimed “46–82% smaller trees with statistically equivalent accuracy”. An audit of every public claim against the code found that the headline figures had been typed into a plotting script as literals, that the remaining figures were stale, that the “equivalence” rested on non-significant cross-fold *t*-tests—a non-significant difference is not evidence of equivalence, and the tests treated non-independent folds as independent (Dietterich, 1998)—and that on the newest genuine run the evolved trees lost to depth-matched CART on 7 of 8 datasets. We withdrew the claims, wrote down three hypotheses and four kill criteria before re-running anything, and report the outcome whichever way it fell (Nosek et al., 2018).

Contributions.

1. A pre-registered, budget-matched evaluation of NSGA-II decision-tree induction on 20 OpenML-CC18 datasets. Evolution outperforms random sampling of the same space (H3 holds); it does not outperform CART’s pruning path (H1 rejected); and its tuned single tree is not equivalent in accuracy to tuned CART (H2 rejected).
2. Evidence on *why* it loses: the evolved frontiers are truncated at small trees, the hypervolume deficit tracks the accuracy that only larger trees reach, and a one-factor-at-a-time ablation rules out the initial small-tree bias as the cause while attributing part of the truncation to held-out selection and part to budget.
3. Robustness checks that leave every verdict unchanged: constraint repair, a CART path capped at the evolved trees’ depth, and an exact optimal sparse-tree solver (GOSDT).
4. An account of the audit: fabricated published figures, and two evaluation-harness errors—selection on the test fold, and a per-fold hypervolume reference point—that each reversed a kill-criterion verdict, in opposite directions.
5. An open harness in which every reported number, table and figure is generated from committed run output, together with the pre-registration, deviation log and all data.

Outline. Section 2 reviews related work, Section 3 describes the algorithm and baselines, and Section 4 the pre-registered design. Section 5 reports the three pre-registered tests, Section 6 the analysis of why the evolutionary frontiers fall short, and Section 7 the robustness checks. Section 8

describes the audit, and Sections 9 and 10 discuss and conclude. Appendix A lists the commands that regenerate every result.

2 Related Work

Evolutionary tree induction. Barros et al. (2012) survey evolutionary algorithms that induce whole decision trees or design their components. Kretowski (2019) presents a framework for global evolutionary induction, Rivera-Lopez et al. (2022) review metaheuristic approaches more broadly, and `evtree` (Grubinger et al., 2014) is an R implementation of single-objective global evolutionary tree induction. Multi-objective variants optimise accuracy against size with Pareto selection (Zhao, 2007). Freitas (2004) argued that weighted formulas, which fix the trade-off between criteria in advance, are largely ad hoc, and that Pareto and lexicographic approaches are more principled. In the comparisons surveyed by Barros et al. (2012), the baselines are greedy inducers and the datasets mostly come from UCI; none of the listed comparators isolates the contribution of the search operators from that of the representation, which is the gap our random-search baseline targets.

Optimal sparse trees. Exact methods find provably optimal trees under a sparsity penalty or depth limit: OCT (Bertsimas and Dunn, 2017) via mixed-integer programming, and OSDT (Hu et al., 2019), GOSDT (Lin et al., 2020), DL8.5 (Aglin et al., 2020) and MurTree (Demirović et al., 2022) via branch-and-bound and dynamic programming over binarised features, made practical on continuous data by threshold guessing (McTavish et al., 2022). Sweeping the penalty yields a frontier directly comparable to an evolved one. The case for evolution against these methods is flexibility rather than optimality: arbitrary, non-decomposable objectives and an anytime answer.

Evaluation methodology. We follow Demšar (2006) in treating datasets as the unit of inference, with Wilcoxon signed-rank tests and Holm correction (Holm, 1979); use nested cross-validation to avoid selection bias (Varma and Simon, 2006; Cawley and Talbot, 2010); test equivalence with two one-sided tests (TOST) (Schuirmann, 1987); and use the Nadeau–Bengio variance correction where folds must be the unit (Nadeau and Bengio, 2003). Bayesian alternatives (Benavoli et al., 2017) answer related questions and could be applied to the committed data. Pre-registration (Nosek et al., 2018) and variance-aware benchmarking (Bouthillier et al., 2021) respond to the same problems we found in our own earlier claims (Lipton and Steinhardt, 2019).

Measuring interpretability. There is no agreed measure of how interpretable a tree is (Lipton, 2018; Doshi-Velez and Kim, 2017). Size- and path-length-based proxies are the most common (Freitas, 2014; Piltaver et al., 2016); we report only those and do not claim that any of them measures interpretability (Rudin, 2019).

3 Method

3.1 Representation and operators

Individuals are binary axis-parallel trees with a maximum depth of 6 and minimum sample counts per split (8) and per leaf (3). Leaves predict the majority class of the training rows reaching them. Initial trees grow recursively, stopping at each node with probability 0.3. Split thresholds are

drawn from the midpoints between consecutive distinct values of the samples reaching the node—the candidate set CART searches—and filtered so both sides satisfy the leaf minimum. Variation uses subtree crossover and four mutations: threshold perturbation snapped to an observed midpoint, feature replacement, subtree pruning and leaf expansion, with weights 0.45, 0.25, 0.25 and 0.05.

3.2 Fitness and selection

Each training fold is split 80/20 (stratified) into a fitting part and a validation part. Leaves are fitted on the former and the tree is scored on the latter, so the search does not reward trees for memorising the rows their leaves were fitted on.

Multi-objective mode. NSGA-II (Deb et al., 2002), using the DEAP implementation (Fortin et al., 2012), optimises (validation accuracy, $-\text{node count}$) with population 80, 40 generations, crossover probability 0.72 and mutation probability 0.18. Structurally identical individuals are removed before environmental selection: crowding distance does not separate clones, and in a preliminary run without this step the number of distinct objective vectors fell from 26 to 2 within five generations. The delivered model set is the final population’s first front.

Single-objective mode. Used for the point-estimate test (Section 5.3): the same representation and operators, tournament selection, elitism of 12%, and a weighted sum of validation accuracy and a composite “interpretability” score. The composite is a search heuristic only; kill criterion K4 (Section 4) forbids reporting it as an outcome. The accuracy weight is tuned by inner cross-validation, and the best individual is delivered.

3.3 Baselines

Random search (RS). Draws trees from the GA’s own initialiser, scores them with the GA’s own fitness and validation split, and keeps a non-dominated archive on those validation objectives. Its budget is set per fold to the number of evaluations the GA actually spent on that fold, read from a counter rather than predicted: NSGA-II’s cost per generation depends on how many offspring variation changes, so the nominal $\text{pop} \times \text{gens}$ (3,200) overstates it. For the single-objective GA, elites carry their fitness across generations, so a run costs $\text{pop} + \text{gens} \times (\text{pop} - n_{\text{elite}}) = 2,920$ evaluations.

Archived GA. Because RS can only deliver an archive, the GA was also run delivering its archive over every candidate it scored, as a control for that bookkeeping difference.

CART pruning path. Scikit-learn CART (Pedregosa et al., 2011) with the same sample minimums, fitted at up to 12 values spread along its cost-complexity pruning path—a frontier obtained from one greedy fit, with no search over whole trees. As pre-registered, the path is not depth-limited, whereas the GA cannot exceed depth 6; Section 7 also reports a depth-capped path.

Point-estimate baselines. For the single-tree test, CART with cost-complexity pruning is tuned by inner CV over depth $\{3, 4, 5, 6, 8\}$ and α on its pruning path; unconstrained CART and a 100-tree random forest are included as references.

3.4 Frontier metric

Every delivered model is scored once on the held-out test fold; each method’s point set is filtered for dominance in (test accuracy, node count), and its hypervolume is the area it dominates up to a reference point of accuracy 0 and one node more than the largest tree any method produced on that dataset (Zitzler and Thiele, 1999). The reference is fixed per dataset, not per fold (Section 8). Dominance filtering after scoring does not change the hypervolume, since dominated points add no area, and every method delivers a set chosen without test labels. For display we divide hypervolume by the reference node count, giving an area fraction comparable across datasets; all tests use the raw values.

4 Pre-Registered Design

The hypotheses, datasets, statistics and kill criteria were written down before any re-run. Deviations were recorded, with their expected direction, before the results they affect were computed (Table 2).

4.1 Hypotheses and kill criteria

- H1** The GA frontier dominates CART’s pruning-path frontier by hypervolume on a majority of datasets.
- H2** Held-out accuracy of the GA’s selected tree is equivalent to inner-CV-tuned CART within ± 2 accuracy points (TOST).
- H3** The GA’s hypervolume exceeds budget-matched random search’s.
- K1** If RS matches the GA (Wilcoxon over datasets, $\alpha = 0.05$), the evolutionary machinery contributes nothing; publish the software only.
- K2** If the GA has the larger hypervolume than CART on fewer than 60% of datasets, H1 is rejected, and may not be softened to “competitive on some datasets”.
- K3** If the GA loses more than 2 points to tuned CART on more than 30% of datasets, no equivalence claim is made. Two readings were fixed before the K3 run: the primary counts datasets whose mean loss exceeds 0.02; the secondary counts datasets on which a per-fold TOST with the Nadeau–Bengio corrected variance fails. Ambiguity between them resolves against the claim.
- K4** The composite “interpretability” score used inside the fitness may not appear as an outcome. Complexity is reported as nodes, leaves, mean decision-path length (over training samples) and distinct features used.

4.2 Datasets

All OpenML-CC18 (Bischl et al., 2021; Vanschoren et al., 2014) classification tasks with 500–1500 rows, at most about 50 features and a smallest class of at least 40: exactly 20 datasets, fixed by rule before any run reported here (Table 1). The rule excludes *iris* and *wine* (fewer than 500 rows), on which most of the withdrawn claims rested; the Wisconsin diagnostic breast-cancer data, the third, is included as *wdbc* because it meets the rule.

Table 1: The pre-registered datasets: all OpenML-CC18 classification tasks meeting the selection rule, in increasing size.

Dataset	OpenML ID	n	p	Classes	Smallest class
dressess_sales	23381	500	12	2	210
kc2	1063	522	21	2	107
climate_crashes	40994	540	18	2	46
wdbc	1510	569	30	2	212
ilpd	1480	583	10	2	167
balance_scale	11	625	4	3	49
credit_approval	29	690	15	2	307
breast_w	15	699	9	2	241
eucalyptus	188	736	19	5	105
blood_transfusion	1464	748	4	2	178
diabetes_pima	37	768	8	2	268
analcata_data_dmft	469	797	4	6	123
vehicle	54	846	18	4	199
tic_tac_toe	50	958	9	2	332
vowel	307	990	12	11	90
credit_g	31	1000	20	2	300
qsar_biodeg	1494	1055	41	2	356
pc1	1068	1109	21	2	77
banknote	1462	1372	4	2	610
pc4	1049	1458	37	2	178

4.3 Evaluation and statistics

The frontier run (H1, H3) uses outer 10×3 repeated stratified cross-validation (30 folds per dataset); per-fold seeds are derived from (dataset, fold, method). The point-estimate run (H2) tunes every method by inner 5-fold CV and uses 10×1 outer CV. Across-dataset comparisons use the Wilcoxon signed-rank test on per-dataset means, Holm-corrected within each family, with Cohen’s d_z as effect size; the four-method Friedman test is reported alongside. H2 is tested by TOST on per-dataset means with a ± 0.02 margin.

5 Results of the Pre-Registered Tests

5.1 K1: NSGA-II versus budget-matched random search

Budgets matched on every fold (0 unequal; mean 2,546 evaluations per fit). The GA’s hypervolume exceeds RS’s by +0.66 on average ($p = 0.011$, Holm $p = 0.032$, $d_z = +0.50$) and is larger on 15 of 20 datasets (Figure 1). **K1 is not triggered; H3 holds.** The archived-GA control does not differ significantly from the standard GA (+0.12, Holm $p = 0.286$), so there is no evidence that the advantage is an artefact of delivering a final population rather than an archive. The four-method Friedman test is not significant ($p = 0.202$; mean ranks GA 2.15, CART 2.25, archived GA 2.70, RS 2.90), so no Nemenyi post-hoc test is licensed. K1 is defined on the pairwise test, which has more power here because two of the four methods are near-duplicates.

Table 2: Deviations from the pre-registration, each recorded before the affected result was computed.

Deviation	Reason and expected direction
Thresholds from observed midpoints, for both searchers	The uniform draw is not CART’s candidate set. Measured beforehand to help RS more than the GA, i.e. to make K1 harder.
Fitness on a 20% validation split, for both searchers	Resubstitution fitness rewards memorisation. Applied identically, so neutral for K1.
Budget from realised evaluation counts	The nominal product under-funds RS; matching realised counts removes a bias toward the GA.
K3 run: 10×1 outer CV	Compute. Wider per-dataset intervals; the across-dataset unit is unchanged.
K3 run: GA and RS tuned over accuracy weight only, at depth 6	Compute. Removes a tuning option from the GA but not from CART: biases K3 against the GA, the conservative side for a criterion that blocks an equivalence claim.
Loader fix for a pandas-version change	Two datasets had been silently re-encoded; afterwards all 20 reproduce the committed CART frontiers exactly.
Constraint repair	Implemented after K1/K2 were known; reported only as a sensitivity analysis (Section 7).

5.2 K2: NSGA-II versus the pruning path

The GA has the larger mean hypervolume than CART’s pruning path on 9 of 20 datasets (45%), below the 60% threshold. **K2 is triggered; H1 is rejected.** The mean difference is -4.21 (Holm $p = 0.368$). Table 3 and Figure 1 show the pattern: the GA wins narrowly on datasets where small trees are near-optimal, and loses heavily on *vowel*, *eucalyptus*, *tic-tac-toe* and *vehicle*, where accuracy keeps rising with tree size.

5.3 K3 and H2: accuracy of a single tuned tree

With its accuracy weight tuned by inner CV, the single-objective GA’s tree loses more than 2 points to tuned CART on 8 of 20 datasets, above the 30% threshold. **K3 is triggered under the primary reading**, and the corrected per-dataset TOST fails on 18 of 20, so the secondary reading agrees. Across datasets the mean difference is -0.038 (90% CI $[-0.069, -0.008]$, TOST $p = 0.843$): the accuracies are not equivalent within ± 2 points. **H2 is rejected, and no equivalence claim is made.** The difference is not significant either (Wilcoxon, Holm $p = 0.072$). This combination—a difference that is neither significant nor shown to lie within the margin—is exactly the one the earlier “equivalence” claim misread as evidence of equivalence.

The losses are concentrated (Table 4, Figure 2). Of the 8 datasets where the GA loses, 8 are among the 8 with the largest hypervolume deficit to CART in the frontier run—a separate run in a different algorithm mode. On the other 12 the GA is within 2 points of CART or ahead of it (ahead by more than 2 points on 1). Mean accuracies are GA 0.738, tuned CART 0.776 and RS 0.726. The GA’s trees have 6.3 leaves (CART 18.4), a mean decision path of 2.24 splits (CART 3.61) and use 2.9 features (CART 6.3). Inner CV chose the most accuracy-favouring weight on offer, 0.9, in 52% of folds, and the trees remain small at that weight. Because the upper end of the grid was chosen in about half the folds, we cannot exclude that a higher accuracy weight would yield larger and more accurate trees. As in K1, the GA also outperforms budget-matched RS on accuracy ($+0.012$, Holm $p = 0.003$).

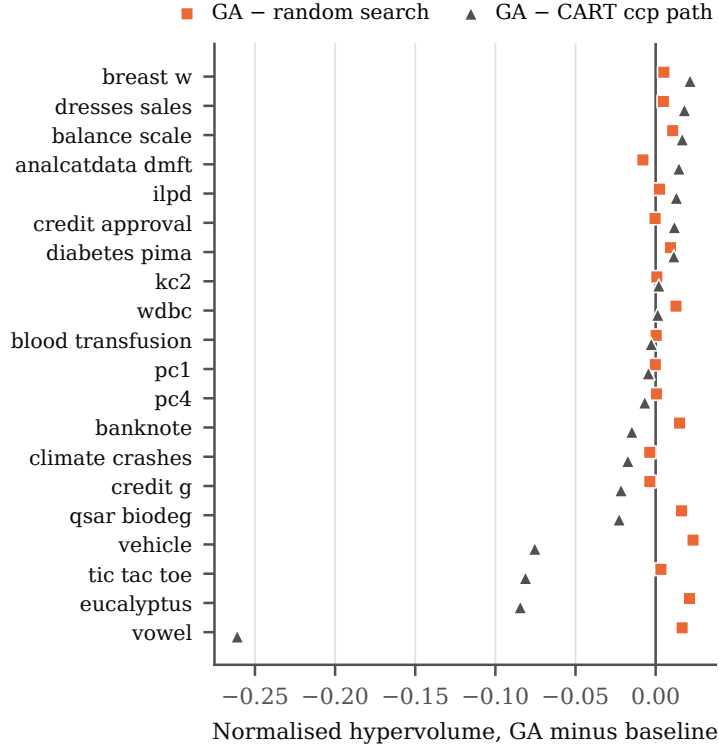


Figure 1: Per-dataset difference in normalised hypervolume between the GA and each baseline (mean over 30 outer folds). Right of zero favours the GA. Datasets are sorted by the difference to CART.

5.4 Summary of verdicts

Criterion	Outcome	Consequence
K1	Not triggered	H3 holds: evolution outperforms random search
K2	Triggered	H1 rejected: no frontier-dominance claim
K3	Triggered	H2 rejected: no equivalence claim
K4	Held	The composite score appears in no reported result

6 Why the Evolved Frontiers Fall Short

This section is exploratory: the analyses were designed after K2 was known and decide nothing.

The frontiers are truncated. The largest tree on the GA’s test front has a median of 5.4 nodes across datasets, against 11.2 for RS and 32.0 for CART, and its best test accuracy averages 0.750 against CART’s 0.801. Across datasets the hypervolume deficit to CART is strongly rank-correlated with that best-accuracy gap (Spearman $\rho = 0.85$, $p < 0.001$). Figure 3 shows one fold of each kind: in both, the GA’s front lies at or above CART’s over the size range it covers, and does not extend to the large trees.

One factor at a time. On the first 10 folds of the 4 datasets where the GA loses most, we changed one thing at a time, with the committed GA as a control arm that reproduces the original

Table 3: Mean normalised hypervolume per dataset (higher is better; best of GA, RS and CART in bold) and the mean node count of the largest tree on the GA’s and CART’s test fronts. GOSDT (Section 7) was run on the first 10 folds with its own reference point, so its column is not on the same scale and is not bolded.

Dataset	GA	RS	CART	GOSDT	Largest tree	
					GA	CART
analcata_data_dmft	0.237	0.245	0.223	0.228	6	94
balance_scale	0.749	0.738	0.732	0.704	13	39
banknote	0.894	0.879	0.909	0.890	13	29
blood_transfusion	0.782	0.781	0.784	0.775	6	32
breast_w	0.906	0.901	0.884	0.861	5	13
climate_crashes	0.892	0.896	0.910	0.895	2	10
credit_approval	0.840	0.840	0.828	0.780	4	22
credit_g	0.723	0.727	0.744	0.729	5	48
diabetes_pima	0.744	0.734	0.732	0.758	6	32
dresses_sales	0.620	0.615	0.602	0.602	5	36
eucalyptus	0.521	0.500	0.606	0.509	12	63
ilpd	0.723	0.721	0.710	0.709	4	13
kc2	0.827	0.826	0.825	0.838	3	13
pc1	0.919	0.919	0.924	0.921	2	16
pc4	0.887	0.886	0.893	0.892	4	23
qsar_biodeg	0.789	0.773	0.812	–	9	45
tic_tac_toe	0.744	0.740	0.825	0.800	10	118
vehicle	0.594	0.570	0.669	0.603	14	77
vowel	0.360	0.343	0.621	0.399	24	155
wdbc	0.899	0.886	0.898	0.829	5	11

fronts exactly (Table 5). Scoring fitness on the fitting data instead of a validation split—so that a larger tree need not prove itself on about a hundred held-out rows—closes 29% of the normalised-hypervolume gap to CART on average. Doubling population and generations closes 28%. Removing the bias toward small trees—seeding full-depth trees and swapping the expand and prune mutation weights—closes −4%, that is, none of it. Under every variant the largest delivered tree stays between 15 and 40 nodes, against CART’s 78–162.

Interpretation. On these datasets the truncation is not an initialisation artefact. Held-out selection and search budget each account for part of it, and most of it is left unexplained by these three factors. By contrast, CART’s path reaches its large trees without scoring them on held-out data and without searching for them. Greedy seeding with CART trees, which would place the large accurate trees in the population directly, is untested, and is the most direct candidate for a follow-up study.

7 Robustness and Exploratory Comparisons

All three checks were added after K1/K2 were known. None can change a verdict, and none does.

Constraint repair. The sample-count constraints were enforced only at initialisation. On one fold of each of 4 datasets, 8–27% of the trees NSGA-II evaluated contained a split the constraints forbid (0% for RS, whose trees all come from the initialiser), yet only 1 of 79 internal nodes in the delivered fronts did, consistent with the size objective removing such splits on its own. We added a

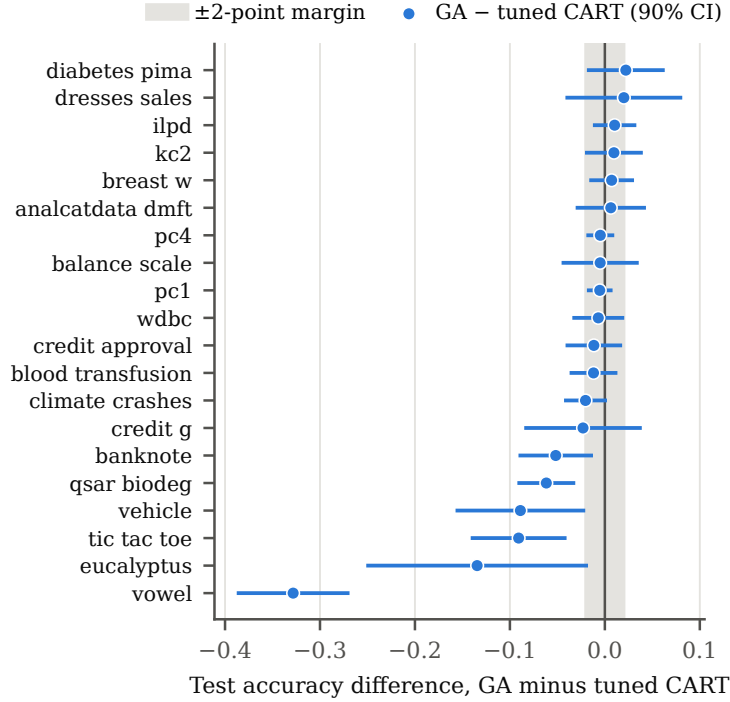


Figure 2: Test-accuracy difference between the tuned GA tree and tuned CART per dataset, with 90% intervals using the Nadeau–Bengio corrected variance over outer folds. The shaded band is the pre-registered ± 2 -point margin.

repair that collapses, top-down, every split reached by fewer than 8 training rows or leaving fewer than 3 on a side. Re-running the frontier benchmark with repair on and everything else fixed gives a GA–RS difference of $+0.85$ (Holm $p = 0.034$, larger on 16/20 datasets), and the GA has the larger hypervolume than CART on 45% of datasets. One secondary result changes: with repair, the standard GA also outperforms its archived variant ($+0.54$, Holm $p = 0.025$), so the choice between delivering the final front and the archive is not neutral once dead splits are removed.

Depth-capped CART path. Re-deriving CART’s path with the GA’s depth cap on the same folds and seeds, the GA has the larger hypervolume on 8 of 20 datasets (40%), so the depth asymmetry does not explain K2.

GOSDT. GOSDT (Lin et al., 2020) with threshold guessing (McTavish et al., 2022), swept over ten regularisation values at the same depth limit and run on the first 10 outer folds, has the larger hypervolume than CART’s path on 7 of the 19 datasets it completed; the GA beats GOSDT on 10. GOSDT did not complete on `qsar_biodeg`, where its search exhausted memory under caps of 8 and 11 GB; that dataset is excluded. It took 14.4s per fold on average; 32 of 1900 fits hit the 30s limit and are therefore not certified optimal, and 10 returned no model and were dropped from the path, which can only lower GOSDT’s hypervolume. An exact sparse-tree solver, in other words, does not dominate CART’s pruning path on these datasets either, and the difference between the GA and GOSDT is not significant (-0.18 , Holm $p = 0.980$). The pruning path is a harder baseline than a single CART tree suggests.

Table 4: Point-estimate run, per dataset: mean test accuracy of the tuned GA and tuned CART, their difference with a 90% Nadeau–Bengio interval, mean leaf counts, and whether the dataset counts against the GA under K3’s primary reading.

Dataset	GA	CART	Δ	90% CI		Leaves GA/CART	Loss >2 pts
vowel	0.382	0.710	−0.328	−0.388	−0.269	20.7 / 60.4	✓
eucalyptus	0.497	0.632	−0.135	−0.251	−0.018	8.7 / 19.7	✓
tic_tac_toe	0.764	0.855	−0.091	−0.141	−0.040	12.4 / 57.0	✓
vehicle	0.604	0.693	−0.089	−0.157	−0.021	11.8 / 26.7	✓
qsar_biodeg	0.761	0.823	−0.062	−0.092	−0.031	5.4 / 17.2	✓
banknote	0.929	0.980	−0.052	−0.091	−0.013	8.1 / 21.0	✓
credit_g	0.700	0.723	−0.023	−0.085	+0.039	3.0 / 11.5	✓
climate_crashes	0.915	0.935	−0.020	−0.043	+0.002	2.0 / 7.3	✓
blood_transfusion	0.763	0.775	−0.012	−0.037	+0.013	2.0 / 11.8	
credit_approval	0.849	0.861	−0.012	−0.041	+0.018	2.9 / 9.7	
wdbc	0.923	0.930	−0.007	−0.034	+0.020	4.2 / 8.1	
pc1	0.928	0.933	−0.005	−0.019	+0.008	3.0 / 9.7	
balance_scale	0.784	0.789	−0.005	−0.046	+0.036	13.3 / 20.3	
pc4	0.890	0.895	−0.005	−0.019	+0.010	2.4 / 8.3	
analcata_data_dmft	0.211	0.204	+0.006	−0.031	+0.043	9.8 / 41.0	
breast_w	0.954	0.947	+0.007	−0.016	+0.031	4.8 / 11.5	
kc2	0.849	0.839	+0.010	−0.021	+0.040	2.8 / 6.4	
ilpd	0.717	0.707	+0.010	−0.013	+0.033	2.2 / 2.9	
dresses_sales	0.598	0.578	+0.020	−0.042	+0.082	4.2 / 4.1	
diabetes_pima	0.742	0.720	+0.022	−0.019	+0.063	3.0 / 12.6	

8 What the Audit Found

The audit behind this study produced five findings that may be of use beyond this system.

Published numbers without runs. The “46–82%” size reduction was spliced from two dictionaries of literals in a plotting script, and a figure titled “Statistical Equivalence to CART” plotted cross-fold *t*-test *p*-values against a hand-set target of 0.55. The fix was structural rather than editorial: the figure generator now reads only committed run output and raises an error when there is none, and every number in this paper is generated from committed files.

Selection on the test fold. The first frontier run reported K1 triggered, with RS ahead by a wide margin on 19 of 20 datasets. RS had been returning every tree it sampled—about 2,500 per fold—and the dominance filter then ran on their test scores: a maximum over thousands of test evaluations that no method could deliver, since choosing among the candidates needs the test labels. The error was visible in the delivered count, 2,545 against the GA’s 10. With both searchers archiving on validation objectives, they deliver 10.2 (GA) and 5.6 (RS) models per fold, and the verdict reverses.

The reference point. The pre-registration fixes the hypervolume reference per dataset; the harness computed it per fold. A smaller box favours whichever method has the lower peak accuracy—here the GA—and the per-fold version put the share of datasets on which the GA has the larger hypervolume at 70%, above the K2 threshold. Computed as specified, it is 45%. The error was

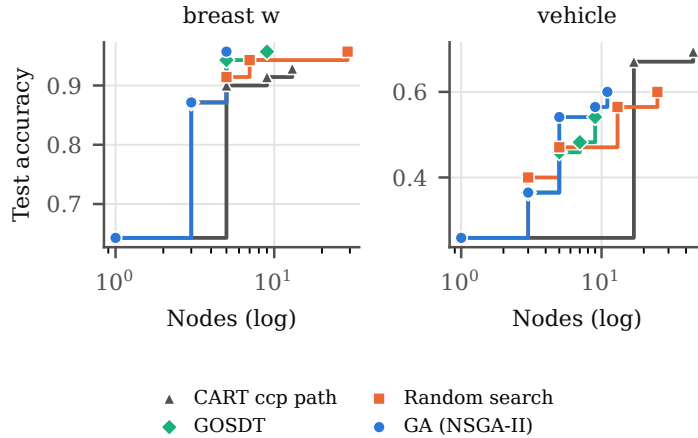


Figure 3: Test-fold frontiers for outer fold 1 of a dataset the GA wins (*breast-w*) and one it loses (*vehicle*).

Table 5: One-factor-at-a-time ablation on the four datasets where the GA loses most (first 10 outer folds): mean normalised hypervolume (HV) and mean node count of the largest delivered tree (Max) per arm. The control arm reproduces the committed GA exactly.

Arm	eucalyptus		tic_tac_toe		vehicle		vowel	
	HV	Max	HV	Max	HV	Max	HV	Max
GA (control)	0.525	11	0.744	13	0.606	14	0.363	25
No validation split	0.537	20	0.773	16	0.637	16	0.384	33
No small-tree bias	0.518	17	0.749	20	0.594	18	0.384	40
2× budget	0.538	15	0.771	15	0.629	16	0.407	27
Random search	0.498	25	0.745	14	0.574	37	0.340	46
CART ccp path	0.608	78	0.820	133	0.661	89	0.619	162

found by reading the implementation against the protocol, not prompted by the result, but it shows that a pre-registration must fix a metric’s parameters and not only its name.

A saved configuration that does not reproduce its run. Each run saved its resolved configuration next to its output. The YAML writer sorted keys alphabetically, and the mutation operator is drawn by position from the ordered weight table, so re-running from the saved file selected different operators for the same seeds and produced different fronts. We found this only because an ablation’s control arm failed to match the committed run; the source configuration reproduces it exactly. Serialisation order is part of the experiment whenever a random draw indexes into a mapping.

Diagnoses that were wrong. Two pre-run screening measurements (three datasets, three folds each, so directional only) contradicted our prior explanations. Midpoint thresholds were adopted to remove a handicap we believed was suppressing the GA; they raised *random search*’s accuracy by 0.012 and lowered the GA’s by 0.003. A plausible explanation is that the GA could already reach good thresholds through threshold mutation over generations, whereas RS, which draws every candidate independently, could not. Lowering the initial stopping probability to 0 to seed bushier trees made the GA lose to RS (−0.018): seeded with full-depth trees, RS keeps the large accurate ones (16.7 leaves) while the GA, whose mutations favour pruning five to one and whose single-

objective fitness rewards small size, ends at 4.4 leaves. Both decisions—keep midpoint thresholds, keep the stopping probability at 0.3—were recorded, with the direction they push K1, before the frontier run.

9 Discussion

What the evidence supports. Over the same tree space and an exactly matched evaluation budget, evolutionary search recovers a better accuracy–size frontier than random sampling, with a medium effect size ($d_z = +0.50$). That is a claim about the search mechanism. It is not a claim about predictive performance relative to standard methods: the evolved frontier does not beat CART’s pruning path, and the tuned single tree is not equivalent in accuracy to tuned CART. What the evolutionary approach retains is controllability—a frontier from one run, and objectives that need not decompose over nodes—which this study neither tests nor refutes.

Recommendations. For the evaluation of evolutionary tree induction we make three recommendations. Report budget-matched random search over the same representation; it is cheap, and without it no advantage can be attributed to evolution. Compare frontiers against the full cost-complexity path, not a single CART tree. And fix the metric’s parameters in the protocol, including reference points and the rule for which models a method delivers: both of our harness errors lived there, and each was large enough to reverse a conclusion.

Threats to validity. *Internal.* The GA was not tuned for the frontier run, and the ablation suggests that a larger budget and fitness without a small validation split each recover part of the gap. We did not pursue either, because tuning the GA after observing K2 would introduce the analytic flexibility the pre-registration is meant to prevent; they are hypotheses for a new pre-registered study. The K3 run used one outer repeat and a reduced grid for the GA, both biasing against it; inner CV chose the upper end of that grid in about half the folds. *External.* The datasets are mid-sized (500–1458 rows, up to 41 features) by design, and a single representation, fitness and budget were tested; other evolutionary tree inducers may behave differently. *Construct.* Hypervolume depends on the reference point, which here depends on the largest tree any method produced on a dataset; the normalised values in the tables are comparable within a dataset, not across studies. Node count is a structural proxy, and nothing here measures whether a person can read these trees faster (Lipton, 2018; Doshi-Velez and Kim, 2017).

10 Conclusion

In this system, evolutionary search outperforms budget-matched random sampling of the same tree space but not CART’s cost-complexity pruning path, which requires no search over whole trees; and its tuned single tree is not as accurate as tuned CART. The shortfall is concentrated on problems where accuracy needs large trees, which the evolved frontiers do not reach. We report this as the main finding, together with the audit trail—the withdrawn claims, the two reversed verdicts and the committed data behind every number—so that the next comparison in this area can start from the baselines that decided this one.

Data and Code Availability

Code, configuration, the pre-registration and its deviation log, and all run output are available at <https://github.com/ibrah5em/ga-optimized-trees> (branch `paper/phase-0`). Appendix A lists the commands that regenerate every result in this paper.

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A Reproducing the Results

All numbers, tables and figures in this paper are generated by `scripts/paper_assets.py` from the committed evidence under `paper/evidence/`; the script leaves a number undefined, so the build fails, when its evidence is missing, and refuses to build if the ablation’s control arm stops reproducing the committed run. The runs themselves are regenerated as follows (datasets are fetched from OpenML).

```
# Frontier run: K1 and K2 (about 35 min on 5 cores)
python scripts/frontier_benchmark.py --config configs/paper.yaml --n-jobs 5
```

```

# Point-estimate run: K3 and H2 (about 2 h on 4 cores; resumable)
python scripts/benchmark.py --config configs/paper.yaml --outer-repeats 1 \
    --no-depth-tuning --n-jobs 4
python scripts/k3_analysis.py paper/evidence/k3-2026-09-29/folds.csv

# Sensitivity and exploratory runs
python scripts/frontier_benchmark.py --config configs/paper-repair.yaml --n-jobs 4
python scripts/cart_depth_matched.py \
    --output-dir paper/evidence/cart-depth6-2026-09-29
python scripts/frontier_diagnostic.py \
    --datasets vowel,eucalyptus,vehicle,tic_tac_toe \
    --output-dir paper/evidence/frontier-diagnostic-2026-09-29
python scripts/gosdt_frontier.py --n-jobs 2 --time-limit 30 --memory-gb 8
python scripts/measure_constraint_violations.py wdbc vehicle credit_g tic_tac_toe \
    --output violations.csv

# Check that locally loaded data matches the committed run
python scripts/verify_dataset_identity.py wdbc banknote

# Regenerate every number, table and figure
python scripts/paper_assets.py

```

Runs must be launched from the configuration files in `configs/`, not from the copies saved next to each run's output (Section 8).